

How Do Hardware and Software Work Together in a Robotic System?

Every robotic system works in similar ways, and they have two types of components: hardware and software. Software is invisible and in robots, the lines of code programmed into it make up software. Hardware is tangible, so it has physical parts to be built into a robotic system. Hardware comprises the sensors like temperature sensors, actuators like motors, circuit wires, and power sources of a robot. For instance, actuators, which function like the muscles of a robot, give output by executing action. Microcontrollers like the Arduino Uno are presented as hardware full of pins to control.

But without software, hardware will only be like a simple toy, having zero interaction with the environment. Software somewhat functions like a human's brain and a neural system; it is responsible for processing incoming information and deciding what to do in return. In a robotic system, it is software that enables it to decide what users see on its face screen. In the case of microcontrollers, software plays a role in its control code.

Similarly, working software depends on the quality of hardware. A robot's software may know it must jump if it encounters a hole in its path, but if it does not have a sense of danger nor the supportive legs to jump from hardware, it will not do what it knows. Other than requiring the visuals' commands from software, a robot's screen has a GPU that renders images, and wires allow it to send the signals around.

Hardware and software depend on each other to create a working robotic system. Sensors are a kind of hardware that picks up signals from the environment. They are converted into electrical signals that travel through circuit wires, which are also part of hardware. The signals reach the controller of the robot, which is physically in the form of hardware with inbuilt control code as software. Control code returns the appropriate response to the signals, so the response signals are sent through the wires to reach the hardware actuators and they do as the robot decides.